EVIDENCE SYNTHESIS FOR PERCUTANEOUS ADHESIOLYSIS IN LOW BACK AND LOWER EXTREMITY PAIN

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Executive Summary:

The growing prevalence of spinal pain in the United States continues to produce substantial economic impact and strain on health-related quality of life. Low back pain, musculoskeletal pain, and neck pain contributed to the top 3 of the 5 conditions leading to the most years lived with disability. Due to its prevalence, the costs of managing spinal pain in the United States continue to increase, costs ranging as high as $100 billion per year. Treatments include a wide array of interventions, starting with simple exercises to complex fusions and repeat operations. Low back and lower extremity pain is often treated with surgery, with the most common indications being disc protrusion with radiculitis, spinal stenosis, and spondylolisthesis. While the evidence supporting surgery for disc protrusion is strong, outcomes for other conditions are less compelling.

Percutaneous adhesiolysis is for multiple recalcitrant, resistant conditions involving spinal pain when epidural injections have failed to adequately manage a patient's pain, especially low back and lower extremity pain, specifically in spinal stenosis and post lumbar surgery syndrome.

There have been multiple randomized controlled trials and systematic reviews with meta-analyses assessing the effectiveness of percutaneous adhesiolysis in chronic low back and lower extremity pain.

In a recent evidence synthesis by Helm et al (1) that included all the available literature on percutaneous adhesiolysis in treating low back and lower extremity pain, significant evidence was shown as a result of a best evidence synthesis utilizing the guidance prepared by the Cochrane review, along with various other agencies including the United States Preventive Services Task Force (USPSTF) and meta-analysis.

Helm et al (1) considered pain relief of at least 50% and functional improvement of at least 40% as the primary outcomes. They identified 7 randomized controlled trials that met the inclusion criteria with appropriate methodologic quality assessment showing efficacy. In fact, they did not find any negative trials. Consequently, they concluded that there is Level I, or strong, evidence for percutaneous adhesiolysis in the treatment of chronic refractory low back and lower extremity pain.

Background:

The growing prevalence of spinal pain in the United States continues to produce substantial economic impact and strain on health-related quality of life. In an assessment by the US Burden of Disease Collaborators for the United States from 1999 to 2010, it was shown that in 2010 low back pain contributed to most years lived with disability (2). The costs of managing spinal pain in the United States, along with its prevalence, continue to increase, ranging up to $100 billion per year. Treatments include a wide array of interventions, starting with simple exercises to complex fusions and repeat operations (3-10). Despite various modalities, specifically surgical interventions, disability and economic burden continue to rise across the globe (2,3,7,8,11).

Chronic refractory low back pain with or without lower extremity pain which does not resolve after conservative therapy or even surgical treatment can present a vexing therapeutic problem. Low back and lower extremity pain is often treated with surgery, with the most common indications for surgery being disc protrusion with radiculitis, spinal stenosis, and spondylolisthesis. While the evidence supporting surgery for disc protrusions is strong, outcomes for other conditions are less compelling. There is a need for therapies to help patients with chronic persistent low back and lower leg pain who either do not want surgery or who are not candidates for surgery. In addition, cost pressures, the introduction of alternate payment systems, and the increasing importance of comparative effectiveness research have all created an impetus for cost-effective therapies, such as interventional techniques, to treat these problems.

Percutaneous adhesiolysis was first described by Racz as a three-day procedure (1,12). The protocol has been modified so that it can be done as a one-day procedure.

There have been multiple systematic reviews assessing the effectiveness of percutaneous adhesiolysis in managing chronic low back and lower extremity pain. Two recent systematic reviews by Helm et al (1) and Manchikanti et al (12) reported significant effectiveness for percutaneous adhesiolysis with Level I to II evidence. In these systematic reviews, rigorous methodologic criteria were utilized, including Cochrane review criteria, Interventional Pain Management Techniques – Quality Appraisal of Reliability and Risk of Bias Assessment (IPM-QRB) (Appendices 1-3), and grading of evidence using 5 levels of evidence ranging from Level I to Level V (Table 1). Quality assessment of the manuscripts and best evidence synthesis, along with meta-analysis were utilized as shown in Appendices 4-6. In their systematic review, Helm et al (1) included 17 randomized controlled trials showing efficacy with no negative results and concluded there is Level I or strong evidence for percutaneous adhesiolysis in the treatment of chronic refractory low back and lower extremity pain. Manchikanti et al (12), assessing only the evidence for post lumbar surgery syndrome, utilized 4 randomized controlled trials which met inclusion criteria. Three of them were high quality and showed efficacy of percutaneous adhesiolysis to be Level II based on a best evidence synthesis.

Table 1. *Grading and synthesis of best evidence by qualitative analysis.*

Level I	Evidence obtained from multiple relevant high quality randomized controlled trials
Level II	Evidence obtained from at least one relevant high quality randomized controlled trial or multiple relevant moderate or low quality randomized controlled trials
Level III	Evidence obtained from at least one relevant moderate or low quality randomized controlled trial with multiple relevant observational studies or Evidence obtained from at least one relevant high quality non-randomized trial or observational study with multiple moderate or low quality observational studies
Level IV	Evidence obtained from multiple moderate or low quality relevant observational studies
Level V	Opinion or consensus of large group of clinicians and/or scientists

Adapted and Modified from: Manchikanti L, et al. A modified approach to grading of evidence. *Pain Physician* 2014; 17:E319-25 (13).

RESULTS OF EVIDENCE ASSESSMENT

Figure 1 shows a study selection flow diagram as recommended by Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) (14).

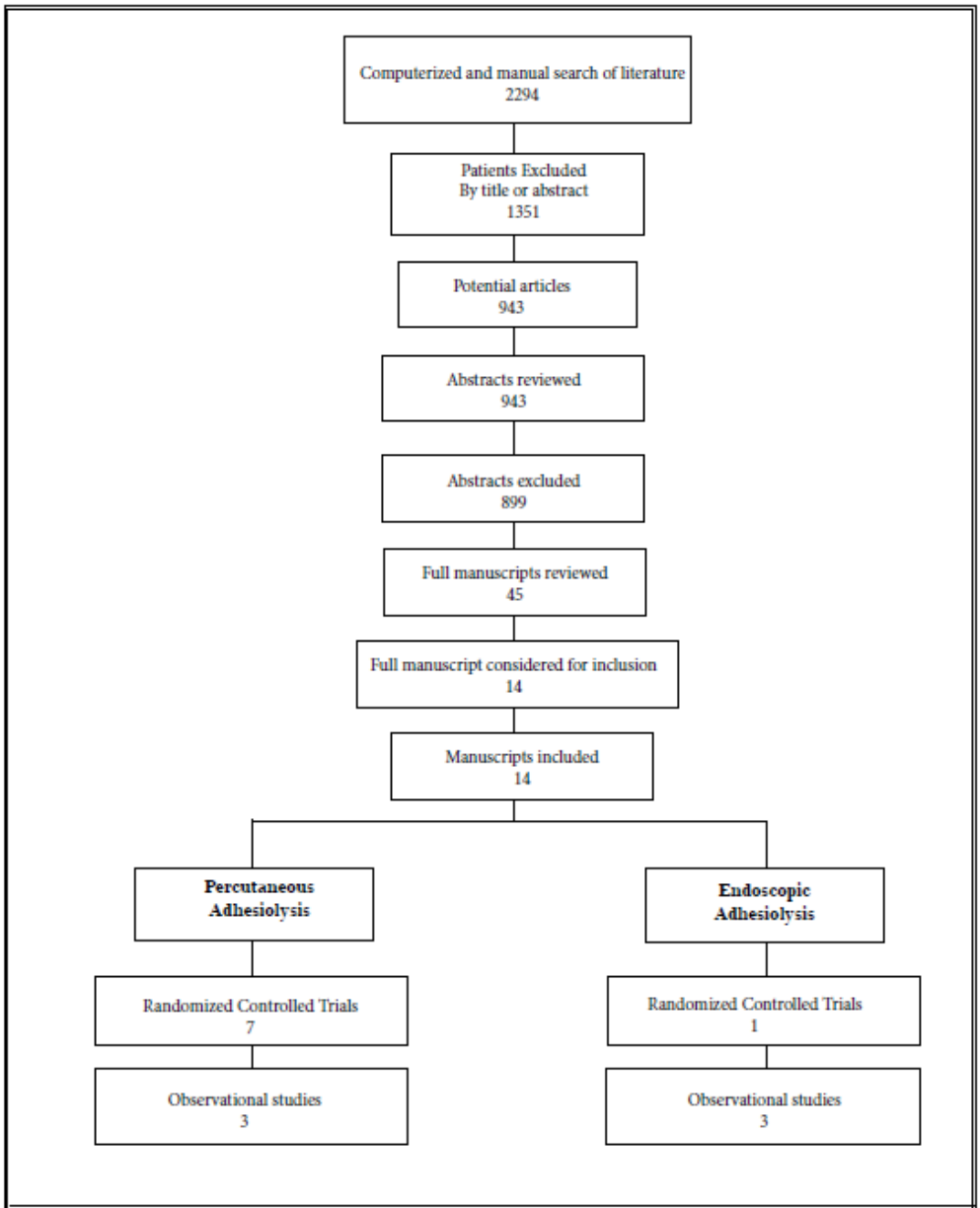


Fig. 1. *Schematic presentation of study selection.*

Appendix 7 illustrates the characteristics of the trials considered for inclusion. There were 7 randomized controlled trials (15-21) and 3 observational trials evaluating percutaneous adhesiolysis (22-24). As there are more than 5 randomized controlled trials, the observational studies are not utilized to assess the quality of evidence.

A meta-analysis was performed for percutaneous adhesiolysis. Studies with no control group or useable numeric data were excluded (21,23,24).

Manchikanti et al (12) concluded there is Level II evidence for post laminectomy syndrome based on 4 randomized trials that met the inclusion criteria for methodological quality assessment; 3 were high quality and the fourth was low quality. Based on these 3 randomized controlled trials, 2 of them reporting on the one-day procedure and one reporting on the 3-day procedure, the level of evidence for the efficacy of percutaneous adhesiolysis is Level II, based on a best evidence synthesis.

Methodological Quality Assessment

A methodological quality assessment of the randomized controlled trials meeting inclusion criteria was carried out utilizing Cochrane review criteria and randomized trials and observational studies utilizing IPM-QRB and Interventional Pain Management Techniques – Quality Appraisal of Reliability and Risk of Bias Assessment for Nonrandomized Studies (IPM-QRBNR) criteria as shown in Appendices 4-6.

Meta-Analysis

There was sufficient homogeneity of the percutaneous adhesiolysis studies to allow a meta-analysis.

The results of the meta-analysis are shown in:

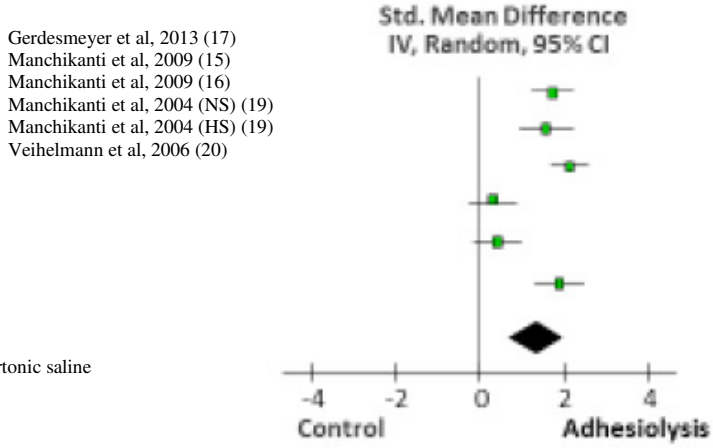
- Table 2, Pain improvement; short-term follow-up of 3 months and long-term follow-up of 12 months,
- Table 3. Functional improvement; short-term follow-up of 3 months and long-term follow-up of 12 months.

An analysis showed there was statistically significant better pain improvement, functional improvement, and successful pain outcome (> 50% pain improvement) for those who underwent percutaneous adhesiolysis than those in the control group for both time points, 3 months and 12 months (Table 4).

Table 2. *Results of meta-analysis of pain status change.*

Short-term follow-up 3 months of pain status.

Study or subgroup	Control			Adhesiolysis			Weight	Std. Mean Difference
	Mean	SD	Total	Mean	SD	Total		IV, random, 95%, CI
Gerdesmeyer et al, 2013 (17)	1.9	1.1	44	3.8	1.1	46	17.1%	1.71 (1.23, 2.20)
Manchikanti et al, 2009 (15)	2.6	1.1	25	4.2	0.9	25	15.9%	1.57 (0.93, 2.21)
Manchikanti et al, 2009 (16)	3	0.8	60	4.7	0.8	60	17.4%	2.11 (1.66, 2.56)
Manchikanti et al, 2004 (NS) (19)	1.2	9.3	25	4	8.9	25	16.6%	0.30 (-0.26, 0.86)
Manchikanti et al, 2004 (HS) (19)	1.2	9.3	25	4.2	4	25	16.5%	0.41 (-0.15, 0.97)
Veihelmann et al, 2006 (20)	0.8	2	27	4.7	2.1	46	16.5%	1.87 (1.30, 2.44)
Total (95% CI)			206			227	100.0%	1.34 (0.71, 1.96)
Heterogeneity: Tau^2 =0.54; Chi^2 =40.88, df=5 (P < 0.00001); I^2 =88%								
Test for overall effect: Z=4.18 (P < 0.00001)								

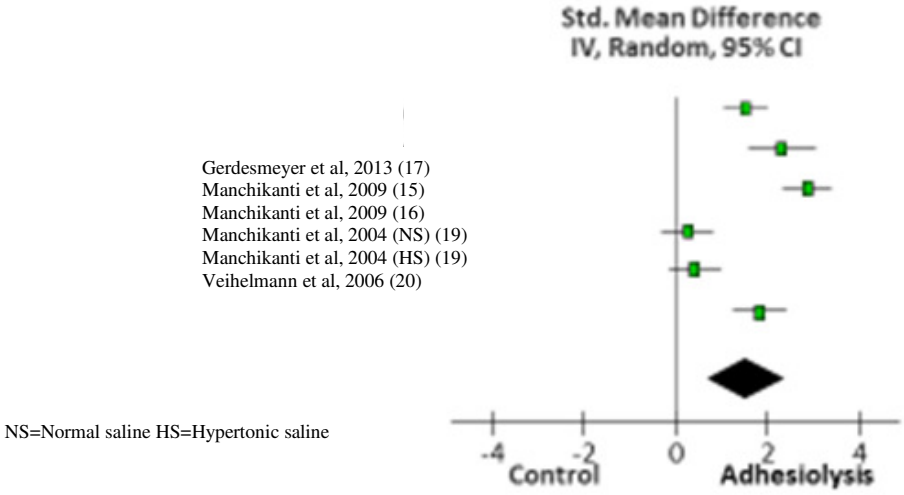


NS=Normal saline HS=Hypertonic saline

Table 2 (cont). *Results of meta-analysis of pain status change.*

Long-term follow-up 12 month of pain status.

Study or subgroup	Control			Adhesiolysis			Weight	Std. Mean Difference
	Mean	SD	Total	Mean	SD	Total		IV, random, 95%, CI
Gerdesmeyer et al, 2013 (17)	3.9	1.1	44	5.5	1	46	17.1%	1.51 (1.04, 1.98)
Manchikanti et al, 2009 (15)	1.8	0.9	25	3.9	0.9	25	15.9%	2.30 (1.57, 3.02)
Manchikanti et al, 2009 (16)	1.8	0.8	60	4.1	0.8	60	16.9%	2.86 (2.34, 3.37)
Manchikanti et al, 2004 (NS) (19)	1.2	9.3	25	3.6	8.9	25	16.7%	0.26 (-0.30, 0.82)
Manchikanti et al, 2004 (HS) (19)	1.2	9.3	25	4.2	4	25	16.7%	0.41 (-0.15, 0.97)
Veihelmann et al, 2006 (20)	0.5	2	27	4.3	2.1	46	16.7%	1.82 (1.26, 2.39)
Total (95% CI)			206			227	100.0%	1.52 (0.70, 2.35)
Heterogeneity: Tau^2 =0.98; Chi^2 =66.25, df=5 (P < 0.00001); I^2 =92%								
Test for overall effect: Z=3.62 (P < 0.0003)								

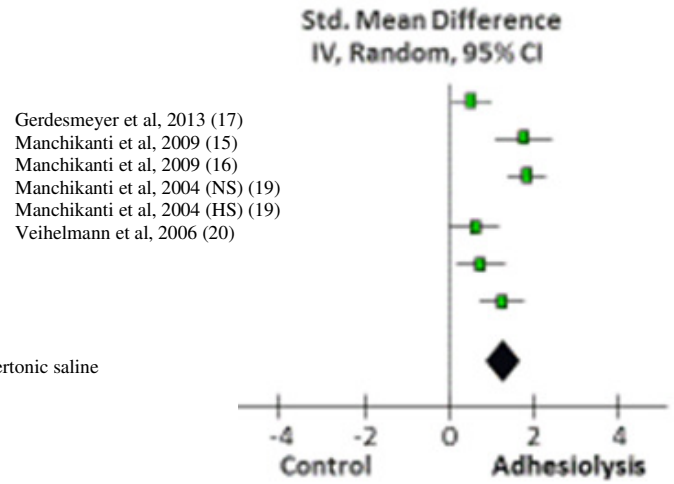


NS=Normal saline HS=Hypertonic saline

Table 3. *Results of meta-analysis of assessment of functional status.*

Functional improvement (short-term follow-up 3 months)

Study or subgroup	Control			Adhesiolysis			Weight	Std. Mean Difference
	Mean	SD	Total	Mean	SD	Total		IV, random, 95%, CI
Gerdesmeyer et al, 2013 (17)	13.6	11.5	44	28.9	10.8	46	18.1%	1.36 (0.90, 1.82)
Manchikanti et al, 2009 (15)	6.9	4.9	25	15	4.1	25	14.4%	1.76 (1.10, 2.43)
Manchikanti et al, 2009 (16)	8.4	4.1	60	16	4.1	60	18.6%	1.84 (1.41, 2.27)
Manchikanti et al, 2004 (NS) (19)	2	14.5	25	11	14.1	25	16.0%	0.62 (0.05, 1.19)
Manchikanti et al, 2004 (HS) (19)	2	14.5	25	12	11.9	25	15.9%	0.74 (0.17, 1.32)
Veihelmann et al, 2006 (20)	3.1	8.1	27	12.5	7.1	46	17.0%	1.24 (0.72, 1.76)
Total (95% CI)			206			227	100.0%	1.27 (0.87, 1.67)
Heterogeneity: Tau^2 =0.17; Chi^2 =17.24, df=5 (P < 0.004); I^2 =71%								
Test for overall effect: Z=6.24 (P < 0.00001)								

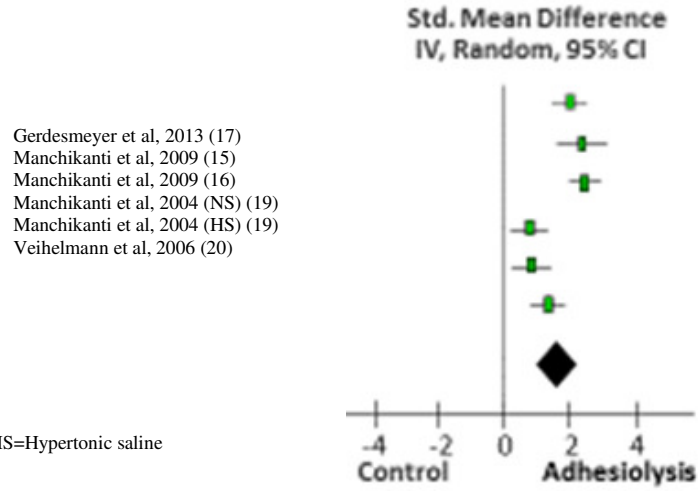


NS=Normal saline HS=Hypertonic saline

Table 3 (cont). *Results of meta-analysis of assessment of functional status.*

Functional improvement (long-term follow-up 12 months)

Study or subgroup	Control			Adhesiolysis			Weight	Std. Mean Difference
	Mean	SD	Total	Mean	SD	Total		IV, random, 95%, CI
Gerdesmeyer et al, 2013 (17)	24.7	11.5	44	45.7	9.3	46	17.2%	2.00 (1.49, 2.51)
Manchikanti et al, 2009 (15)	4.8	4.4	25	15	4.1	25	15.1%	2.36 (1.63, 3.10)
Manchikanti et al, 2009 (16)	5.3	4.1	60	15.4	4.1	60	17.5%	2.45 (1.97, 2.92)
Manchikanti et al, 2004 (NS) (19)	2	14.1	25	13	14.1	25	16.6%	0.77 (0.19, 1.34)
Manchikanti et al, 2004 (HS) (19)	2	14.1	25	13	11.9	25	16.6%	0.83 (0.25, 1.41)
Veihelmann et al, 2006 (20)	0.2	8.1	27	11.6	8.7	46	17.1%	1.33 (0.80, 1.85)
Total (95% CI)			206			227	100.0%	1.62 (1.02, 2.21)
Heterogeneity: Tau^2 =0.47; Chi^2 =34.25, df=5 (P < 0.00001); I^2 =85%								
Test for overall effect: Z=5.33 (P < 0.00001)								

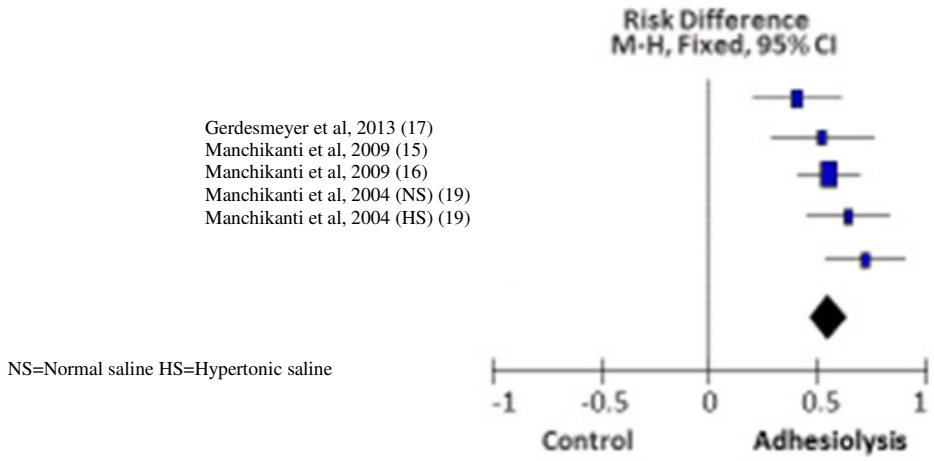


NS=Normal saline HS=Hypertonic saline

Table 4. *Results of meta-analysis of significant improvement ≥50% pain relief.*

Successful outcomes for pain (> 50% pain improvement) (short-term follow-up 3 months)

Study or subgroup	Control		Adhesiolysis		Weight	Risk Difference
	Events	Total	Events	Total		M-h, Fixed, 95%, CI
Gerdesmeyer et al, 2013 (17)	12	36	31	42	22.3%	0.40 (0.20, 0.61)
Manchikanti et al, 2009 (15)	7	25	20	25	14.4%	0.52 (0.28, 0.76)
Manchikanti et al, 2009 (16)	21	60	54	60	34.5%	0.55 (0.41, 0.69)
Manchikanti et al, 2004 (NS) (19)	0	25	16	25	14.4%	0.64 (0.45, 0.83)
Manchikanti et al, 2004 (HS) (19)	0	25	18	25	14.4%	0.72 (0.54, 0.90)
Total (95% CI)		171		177	100.0%	0.55 (0.47, 0.64)
Heterogeneity: Chi^2 =6.19, df=4 (P = 0.19); I^2 =35%						
Test for overall effect: Z=12.72 (P < 0.00001)						

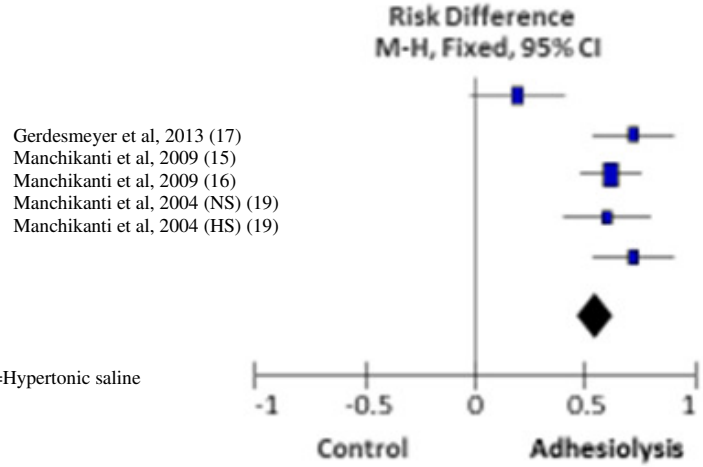


NS=Normal saline HS=Hypertonic saline

Table 4 (cont). *Results of meta-analysis of significant improvement ≥ 50% pain relief.*

Successful outcomes for pain (> 50% pain improvement) (long-term follow-up 12 months)

Study or subgroup	Control		Adhesiolysis		Weight	Risk Difference
	Events	Total	Events	Total		M-h, Fixed, 95%, CI
Gerdesmeyer et al, 2013 (17)	18	36	29	42	22.3%	0.19 (0.02, 0.41)
Manchikanti et al, 2009 (15)	1	25	19	25	14.4%	0.72 (0.54, 0.90)
Manchikanti et al, 2009 (16)	7	60	44	60	32.5%	0.62 (0.40, 0.75)
Manchikanti et al, 2004 (NS) (19)	0	25	15	25	14.4%	0.60 (0.40, 0.80)
Manchikanti et al, 2004 (HS) (19)	0	25	18	25	14.4%	0.72 (0.54, 0.90)
Total (95% CI)		171		177	100.0%	0.55 (0.47, 0.63)
Heterogeneity: Chi^2 =18.57, df=4 (P = 0.001); I^2 =78%						
Test for overall effect: Z=12.97 (P < 0.00001)						



NS=Normal saline HS=Hypertonic saline

Study Characteristics

Appendix 8 shows the characteristics of the included randomized trials and observational studies evaluating percutaneous adhesiolysis.

There were 7 randomized controlled trials dealing with percutaneous adhesiolysis. Gerdesmeyer et al (17) used a 3-day protocol with a ventrolateral catheter placement to evaluate patients with radicular pain and no muscle weakness, with 28 of 31 patients having more than a 50% improvement in the Oswestry Disability Index.

Chun-jing et al (18) looked at patients with radiculopathy after surgery, with a 3.24 mean decrease in the visual analog scale at 6 months.

Manchikanti et al, in 2 different studies (15,16), one looking at patients with post lumbar surgery syndrome(16) and another looking at patients with spinal stenosis (15), found that over 70% of both groups had clinically significant improvements in pain and function.

In a heterogeneous population, Heavner et al (21) found that adhesiolysis provided pain relief.

Manchikanti et al (19) found that both hypertonic saline and normal saline provided clinically significant relief.

Veihelmann et al (20) found clinically significant improvement in pain and function at up to 12 months.

Percutaneous Adhesiolysis

There were 7 randomized controlled trials (15-21) and 3 observational trials evaluating percutaneous adhesiolysis (22-24) showing efficacy or effectiveness for percutaneous adhesiolysis (Table 5). Since the publication of the 2012 systematic review of percutaneous adhesiolysis, there have been 2 new high-quality randomized controlled trials: one from Kiel,

Germany and one from Beijing (17,18). There are now multiple indications for which adhesiolysis has shown to be efficacious, including low back and radicular pain without radiculopathy (17,20); low back and radicular pain with radiculopathy (18); post lumbar surgery syndrome (16); spinal stenosis (15); and low back and leg pain with unspecified radiculopathy or origin (19,21). All 7 studies were randomized controlled trials.

Table 5. *Analysis of effectiveness of percutaneous adhesiolysis.*

Study Study Characteristic Methodological Quality Scoring	Patients	Interventions	Pain Relief and Function			Result		Comments
			3 mos.	6 mos.	12 mos.	Short-Term <6 mos.	Long-Term ≥ 6 mos.	
Gerdesmeyer et al, 2013 (17)	90	Ventral adhesiolysis, 3-day	26/45 of treated group had >50% improvement in ODI	31/42 of treated group had >50% improvement in ODI	28/31 of treated group had >50% improvement in ODI	P	P	High quality placebo controlled study documenting efficacy
Chun-jing et al, 2012 (18)	76	Ventral adhesiolysis with high volume saline	NA	>3 points relief on VAS, with 46% relief	NA	P	NA	High quality study documenting efficacy with high volume saline.
Manchikanti et al, 2009 (16)	120	One-day adhesiolysis v caudal ESI	58% of adhesiolysis had >50% relief vs 38% of comparator	54% of adhesiolysis had >50% relief vs 27% of comparator	51% of adhesiolysis had >50% relief vs 23% of comparator	P	P	High quality study showing efficacy with repeated one-day adhesiolysis procedure in post lumbar surgery syndrome
Heavner et al, 1999 (21)	Adhesiolysis with 0.9% or 10% saline and with or without hyaluronidase	3-day adhesiolysis with either 0.9% or 10% saline and with or without hyaluronidase	About 50% of subjects had more than 10/100 improvement in VAS	About 50% of subjects had more than 10/100 improvement in VAS	About 50% of subjects had more than 10/100 improvement in VAS	P	P	Moderate quality study showing equivalency between 0.9% and 10% saline and with and without saline
Manchikanti et al, 2009 (15)	50	One-day adhesiolysis v caudal ESI	80% of adhesiolysis had >50% relief vs 26% for caudal	80% of adhesiolysis had >50% relief vs 12% for caudal	76% of adhesiolysis had >50% relief at 12 months after 3.5 average injections	P	P	High quality study showing efficacy with repeated one-day adhesiolysis procedure in spinal stenosis patients
Manchikanti et al, 2004 (19)	75	One day adhesiolysis with 0.9% and 10% saline versus epidural injection	72% of 10% saline group, 64% of 0/9% group and 0% of caudal had >50% relief.	72% of 10% saline group, 60% of 0/9% group and 0% of caudal had >50% relief.	72% of 10% saline group, 60% of 0/9% group and 0% of caudal had >50% relief.	P	P	High quality study showing equivalency between normal and hypertonic saline adhesiolysis.
Veihelmann et al, 2006 (20)	99	One-day adhesiolysis with	Mean improvement of	Mean improvement of	Mean improvement of	P	P	Moderate quality study showing superiority of

		10% saline vs. physical therapy	the treated group was >50% in VAS and >40% in ODI. Treatment group had ~10% relief.	the treated group was >50% in VAS and >40% in ODI. Treatment group had ~10% relief.	the treated group was >50% in VAS and >40% in ODI. Treatment group had ~10% relief.			adhesiolysis over physical therapy in persistent radicular pain
Manchikanti et al, 1999 (22)	150	1 vs 2 vs 3 day adhesiolysis. Three-day results from a different study	Average of 37% of 2-day adhesiolysis and 26% had >50% relief after 4 procedures	Average of 21% of 2-day adhesiolysis and 14% had >50% relief after 4 procedures.	Average of 4% of 2-day adhesiolysis and 4% had >50% relief	P	P	No differences between 1,2 and 3 day procedures
Gerdesmeyer et al, 2005 (23)	61	3-day ventral adhesiolysis	>50% improvement in ODI	>50% improvement in ODI	NA	P	P	No control Heterogeneous population
Oh et al, 2014 (24)	303	One-day adhesiolysis comparing ventral with dorsal catheter placement4	Statically significant improvement in leg pain in ventral vs. dorsal group. Ventral and dorsal groups both had >50% relief for back and leg pain.	Statically significant improvement in leg pain in ventral vs. dorsal group. Ventral and dorsal groups both had >50% relief for back and leg pain.	NA	P	P	No control Moderate quality retrospective study

ESI = epidural steroid injection; ODI = Oswestry Disability Index; VAS = Visual Analog Scale; P = positive; NA = not applicable

Based upon the grading of best evidence synthesis, as shown in Table 1, there is Level I or strong evidence for the efficacy of percutaneous adhesiolysis in the treatment of low back and radicular pain.

Complications

Complications of percutaneous epidural adhesiolysis and endoscopic adhesiolysis have been extensively reviewed (25-36).

The most commonly noted complication of percutaneous adhesiolysis has been dural puncture. Veihelmann et al (20) noted 2 instances of dural puncture out of 47 patients. The greatest concern with dural puncture is the risk of the spread of hypertonic saline into the subarachnoid space (37). Generally, as long as hypertonic saline does not get into the subarachnoid space, dural puncture does not require treatment and is not of significant clinical concern. There has been one

report of decreased cerebrospinal fluid pressure due to a chronic subdural hematoma after dural puncture (38).

Transient neurologic deficits have been reported. Generally, these deficits are spontaneously resolved (20,39). A case report of arachnoiditis has been presented, but the volumes injected were very high and the injections were done despite the patient's complaints of unexpected pain (30).

Lee et al (40) reported a rare complication, reverse Takotsubo cardiomyopathy, after the subarachnoid injection of hypertonic (10%) saline during lumbar adhesiolysis. Reverse Takotsubo cardiomyopathy is a variant of stress-induced cardiomyopathy, in which stress causes cardiomyopathy.

Birkenmaier et al (41) performed an in vitro evaluation of the effects of the medications used in adhesiolysis on fibroblast proliferation. This study was prompted by a case of urinary incontinence after adhesiolysis, with no known subarachnoid injection. In culture, fibroblast proliferation was reduced. Hypertonic saline inhibits scar formation because of the effects of human fibrocytes. The relationship of fibroblast inhibition and urinary incontinence is not clear.

Catheter shearing has also been reported (17,27,29). With the change in type of needle used to enter the vertebral foramen away from the RK needle, the risk of shearing is essentially removed. Unless symptomatic, the catheter is left in situ.

As with any procedure, there is a risk of infection or hematoma (22,23,25,26,42). A needle point in the buttock, away from the intertriginous region, is an important safety technique.

No cases of epidural hematoma have been reported.

There are no reported cases of serious neurologic deficits after adhesiolysis, other than the one case of urinary incontinence, described above. The incidence of complications from percutaneous adhesiolysis is low and the complications are generally minimal and self-limited.

DISCUSSION

Refractory low back and lower extremity pain has been successfully treated with percutaneous adhesiolysis. Studied indications include low back and leg pain with radicular symptoms in the absence of motor weakness, spinal stenosis, and post lumbar surgery syndrome. As its name implies, adhesiolysis is primarily focused on breaking up adhesions. However, the wide variety of conditions treated suggest that other mechanisms may be at play (1,12,17,43-49). One of these other mechanisms might be the peridural membrane, which is innervated and can become inflamed, particularly in the infraradicular space, causing low back pain without scarring. The lack of correlation with the extent of central spinal stenosis and relief from adhesiolysis supports the role that factors other than scarring are being relieved by adhesiolysis, as does the fact that the catheter is not moved during the three-day adhesiolysis procedure.

Some question whether scarring can cause pain because of the presence of scarring after successful lumbar surgery (43-45). However, there is extensive literature suggesting that scarring can be painful (46-48).

Epidural fibrosis can cause pain via several mechanisms. One is tethering of the nerve root, so that the root is no longer able to move freely in the epidural space and foramen with movement of the body (49). A trapped nerve root is susceptible to tension and compression and lack of nutrition. Circulation can be impaired causing ischemic pain. Veins can become engorged, resulting in either impaired circulation or direct compression of the nerve. Such congestion can be caused by outflow obstruction, arteriovenous anastomoses or inflammation with secondary congestion. Improvement in filling defects after adhesiolysis is associated with pain relief. Kuslich et al (50) identified scarring between the posterior longitudinal ligament and

the ventral dura, but did not highlight that scarring is a source of pain because he could not differentiate between pain from that scarring and pain from the disc annulus.

Epidural fibrosis is usually diagnosed with magnetic resonance imaging or computed tomography scan; they both have sensitivity and specificity of about 50%-70% for identifying dorsal and lateral epidural scarring (51).

Racz et al (49) developed a clinical test for epidural fibrosis, particularly ventral adhesions, known as the "dural tug." The test consists of the patient sitting with the legs extended then flexing at the waist until back pain first occurs followed by rapidly flexing the neck. Patients are remarkably able to identify their pain level accurately.

Birkenmaier et al (52) performed in vitro studies confirming that dense, post-surgical scarring would not be mechanically lysed. This finding conforms with the clinical experience.

Adhesiolysis depends upon the placement of a spring-wound catheter in the correct tissue plane, in the ventrolateral aspect of the foramen, to allow injection of radiopaque dye, local anesthetic, steroids, saline and hyaluronidase (52). Both 0.9 (normal) saline and 10%, (hypertonic) saline, are effective (19). The procedure is clearly distinct from an epidural steroid injection.

One aspect of adhesiolysis which is often overlooked is post procedural exercises to stretch the nerve root, called "neural flossing." These add mechanical traction to the epidural hydrostatic lysis.

Percutaneous adhesiolysis has a high cost utility at about $2,650 for one quality-adjusted life year for both central spinal stenosis and post lumbar surgery syndrome (53).

There are now 7 RCTs, with supportive observational studies, documenting the efficacy of adhesiolysis. There are no RCTs which do not show efficacy. A meta-analysis confirms the

efficacy of percutaneous adhesiolysis to decrease pain and improve function. The indications for which these RCTs have been done vary widely and include post lumbar surgery syndrome, spinal stenosis, and radicular pain with a disc bulge and without motor weakness. The appropriate indication for percutaneous adhesiolysis is low back and leg pain refractory to conservative treatment, including epidural steroid injections. Gerdesmeyer et al's (17) recommendation that percutaneous adhesiolysis should be the first line treatment for patients with chronic lumbosacral radicular pain should be accepted.

CONCLUSION

Percutaneous adhesiolysis of epidural adhesions to treat refractory low back and lower extremity pain is a technique whose efficacy has been documented by multiple randomized controlled trials. It is a safe and effective procedure, with minimal complications when performed by trained practitioners. A meta-analysis documents its efficacy for pain relief and improved function.

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Appendix 1. *Sources of risk of bias and Cochrane Review rating system.*

A	1. Was the method of randomization adequate?	Yes/No/Unsure
B	2. Was the treatment allocation concealed?	Yes/No/Unsure
C	Was knowledge of the allocated interventions adequately prevented during the study?	
	3. Was the patient blinded to the intervention?	Yes/No/Unsure
	4. Was the care provider blinded to the intervention?	Yes/No/Unsure
	5. Was the outcome assessor blinded to the intervention?	Yes/No/Unsure
D	Were incomplete outcome data adequately addressed?	
	6. Was the drop-out rate described and acceptable?	Yes/No/Unsure
	7. Were all randomized participants analysed in the group to which they were allocated?	Yes/No/Unsure
E	8. Are reports of the study free of suggestion of selective outcome reporting?	Yes/No/Unsure
F	Other sources of potential bias:	
	9. Were the groups similar at baseline regarding the most important prognostic indicators?	Yes/No/Unsure
	10. Were co-interventions avoided or similar?	Yes/No/Unsure
	11. Was the compliance acceptable in all groups?	Yes/No/Unsure
	12. Was the timing of the outcome assessment similar in all groups?	Yes/No/Unsure

Source: Furlan AD, et al. 2009 updated method guidelines for systematic reviews in the Cochrane Back Review Group. *Spine (Phila Pa 1976)* 2009; 34:1929-1941 (54).

Appendix 2. *Item checklist for assessment of randomized controlled trials of IPM techniques utilizing IPM – QRB.*

		Scoring
I.	**TRIAL DESIGN AND GUIDANCE REPORTING**	
1.	**CONSORT or SPIRIT**	
	Trial designed and reported without any guidance	0
	Trial designed and reported utilizing minimum criteria other than CONSORT or SPIRIT criteria or trial was conducted prior to 2005	1
	Trial implies it was based on CONSORT or SPIRIT without clear description with moderately significant criteria for randomized trials or the trial was conducted before 2005	2
	Explicit use of CONSORT or SPIRIT with identification of criteria or trial conducted with high level reporting and criteria or conducted before 2005	3
II.	**DESIGN FACTORS**	
2.	**Type and Design of Trial**	
	Poorly designed control group (quasi selection, convenient sampling)	0
	Proper active-control or sham procedure with injection of active agent	2
	Proper placebo control (no active solutions into active structures)	3
3.	**Setting/Physician**	
	General setting with no specialty affiliation and general physician	0
	Specialty of anesthesia/PMR/neurology/radiology/ortho, etc.	1
	Interventional pain management with interventional pain management physician	2
4.	**Imaging**	
	Blind procedures	0
	Ultrasound	1
	CT	2
	Fluoro	3
5.	**Sample Size**	
	Less than 50 participants in the study without appropriate sample size determination	0
	Sample size calculation with less than 25 patients in each group	1
	Appropriate sample size calculation with at least 25 patients in each group	2
	Appropriate sample size calculation with 50 patients in each group	3
6.	**Statistical Methodology**	
	None or inappropriate	0
	Appropriate	1
III.	**PATIENT FACTORS**	
7.	**Inclusiveness of Population**	
7a.	For epidural procedures:	
	Poorly identified mixed population	0
	Clearly identified mixed population	1
	Disorders specific trials (i.e. well defined spinal stenosis and disc herniation, disorder specific, disc herniation or spinal stenosis or post surgery syndrome)	2
7b.	For facet or sacroiliac joint interventions:	
	No diagnostic blocks	0
	Selection with single diagnostic blocks	1
	Selection with placebo or dual diagnostic blocks	2
8.	**Duration of Pain**	
	Less than 3 months	0
	3 to 6 months	1
	> 6 months	2
9.	**Previous Treatments**	
	Conservative management including drug therapy, exercise therapy, physical therapy, etc.	
	Were not utilized	0
	Were utilized sporadically in some patients	1
	Were utilized in all patients	2
10.	**Duration of Follow-up with Appropriate Interventions**	
	Less than 3 months or 12 weeks for epidural or facet joint procedures, etc., and 6 months for intradiscal procedures and implantables	0
	3 to 6 months for epidural or facet joint procedures, etc., or 1 year for intradiscal procedures or implantables	1
	6 months to 17 months for epidurals or facet joint procedures, etc., and 2 years or longer for discal procedures and implantables	2
	18 months or longer for epidurals and facet joint procedures, etc., or 5 years or longer for discal procedures and implantables	3
IV.	**OUTCOMES**	
11.	Outcomes Assessment Criteria for Significant Improvement	
	No descriptions of outcomes OR	0

	< 20% change in pain rating or functional status	
	Pain rating with a decrease of 2 or more points or more than 20% reduction OR functional status improvement of more than 20%	1
	Pain rating with decrease of ≥ 2 points AND ≥ 20% change or functional status improvement of ≥ 20%	2
	Pain rating with a decrease of 3 or more points or more than 50% reduction OR functional status improvement with a 50% or 40% reduction in disability score	2
	Significant improvement with pain and function ≥ 50% or 3 points and 40% reduction in disability scores	4
12.	**Analysis of All Randomized Participants in the Groups**	
	Not performed	0
	Performed without intent-to-treat analysis without inclusion of all randomized participants	1
	All participants included with or without intent-to-treat analysis	2
13.	**Description of Drop Out Rate**	
	No description of dropouts, despite reporting of incomplete data or ≥ 20% withdrawal	0
	Less than 20% withdrawal in one year in any group	1
	Less than 30% withdrawal at 2 years in any group	2
14.	**Similarity of Groups at Baseline for Important Prognostic Indicators**	
	Groups dissimilar with significant influence on outcomes with or without appropriate randomization and allocation	0
	Groups dissimilar without influence on outcomes despite appropriate randomization and allocation	1
	Groups similar with appropriate randomization and allocation	2
15.	**Role of Co-Interventions**	
	Co-interventions were provided but were not similar in the majority of participants	0
	No co-interventions or similar co-interventions were provided in the majority of the participants	1
V.	**RANDOMIZATION**	
16.	**Method of Randomization**	
	Quasi randomized or poorly randomized or not described	0
	Adequate randomization (coin toss, drawing of balls of different colors, drawing of ballots)	1
	High quality randomization (computer generated random sequence, pre-ordered sealed envelopes, sequentially ordered vials, telephone call, pre-ordered list of treatment assignments, etc.)	2
VI.	**ALLOCATION CONCEALMENT**	
17.	**Concealed Treatment Allocation**	
	Poor concealment of allocation (open enrollment) or inadequate description of concealment	0
	Concealment of allocation with borderline or good description of the process with probability of failure of concealment	1
	High quality concealment with strict controls (independent assignment without influence on the assignment sequence)	2
VII.	**BLINDING**	
18.	**Patient Blinding**	
	Patients not blinded	0
	Patients blinded adequately	1
19.	**Care Provider Blinding**	
	Care provider not blinded	0
	Care provider blinded adequately	1
20.	**Outcome Assessor Blinding**	
	Outcome assessor not blinded or was able to identify the groups	0
	Performed by a blinded independent assessor with inability to identify the assignment-based provider intervention (i.e., subcutaneous injection, intramuscular distant injection, difference in preparation or equipment use, numbness and weakness, etc.)	1
VIII.	**CONFLICTS OF INTEREST**	
21.	**Funding and Sponsorship**	
	Trial included industry employees	-3
	Industry employees involved; high levels of funding with remunerations by industry or an organization funded with conflicts	-3
	Industry or organizational funding with reimbursement of expenses with some involvement	0
	Industry or organization funding of expenses without involvement	1
	Funding by internal resources only with supporting entity unrelated to industry	2
	Governmental funding without conflict such as NIH, NHS, AHRQ	3
22.	**Conflicts of Interest**	
	None disclosed with potential implied conflict	0
	Marginally disclosed with potential conflict	1
	Well disclosed with minor conflicts	2
	Well disclosed with no conflicts	3
	Hidden conflicts with poor disclosure	–1

	Misleading disclosure with conflicts	–2
	Major impact related to conflicts	–3
TOTAL MAXIMUM		**48**

Source: Manchikanti L, et al. Assessment of methodologic quality of randomized trials of interventional techniques: Development of interventional pain management specific instrument. *Pain Physician* 2014; 17:E263-E290 (55).

Appendix 3. *IPM checklist for assessment of nonrandomized or observational studies of IPM techniques utilizing IPM-QRBNR.*

I.	STUDY DESIGN AND GUIDANCE REPORTING	Scoring
1.	**STROBE or TREND Guidance**	
	Case Report/Case Series	0
	Study designed without any guidance	1
	Study designed with minimal criteria and reporting with or without guidance	2
	Study designed with moderately significant criteria or implies it was based on STROBE or TREND without clear description or the study was conducted before 2011 or similar criteria utilized with study conducted before 2011	3
	Designed with high level criteria or explicitly uses STROBE or TREND with identification of criteria or conducted prior to 2011	4
II.	**DESIGN FACTORS**	
2.	**Study Design and Type**	
	Case report or series (uncontrolled – longitudinal)	0
	Retrospective cohort or cross-sectional study	1
	Prospective cohort case-control study	2
	Prospective case control study	3
	Prospective, controlled, nonrandomized	4
3.	**Setting/Physician**	
	General setting with no specialty affiliation and general physician	0
	Specialty of anesthesia/PMR/neurology, etc.	1
	Interventional pain management with interventional pain management physician	2
4.	**Imaging**	
	Blind procedures	0
	Ultrasound	1
	CT	2
	Fluoro	3
5.	**Sample Size**	
	Less than 100 participants without appropriate sample size determination	0
	At least 100 participants in the study without appropriate sample size determination	1
	Sample size calculation with less than 50 patients in each group	2
	Appropriate sample size calculation with at least 50 patients in each group	3
	Appropriate sample size calculation with 100 patients in each group	4
6.	**Statistical Methodology**	
	None	0
	Some statistics	1
	Appropriate	2
III.	**PATIENT FACTORS**	
7.	**Inclusiveness of Population**	
7a.	For epidural procedures:	
	Poorly identified mixed population	1
	Poorly identified mixed population with large sample (≥ 200)	2
	Clearly identified mixed population	3
	Disorders specific trials (i.e. well defined spinal stenosis and disc herniation, disorder specific, disc herniation or spinal stenosis or post surgery syndrome)	4
7b.	For facet or sacroiliac joint interventions:	
	No specific selection criteria	1
	No diagnostic blocks based on clinical symptomatology	2
	Selection with single diagnostic blocks	3
	Selection with placebo or dual diagnostic blocks	4
8.	**Duration of Pain**	
	Less than 3 months	0
	3 to 6 months	1
	> 6 months	2
9.	**Previous Treatments**	
	Conservative management including drug therapy, exercise therapy, physical therapy, etc.	
	Were not utilized	0
	Were utilized sporadically in some patients	1
	Were utilized in all patients	2
10.	**Duration of Follow-up with Appropriate Interventions**	
	Less than 3 months or less for epidural or facet joint procedures, etc., and 6 months for intradiscal procedures and implantables	1
	3-6 months for epidural or facet joint procedures, etc., or one year for intradiscal procedures or implantables	2
	6-12 months for epidurals or facet joint procedures, etc., and 2 years or longer for discal procedures and implantables	3
	18 months or longer for epidurals and facet joint procedures, etc., or 5 years or longer for discal procedures and implantables	4

IV.	**OUTCOMES**	
11.	**Outcomes Assessment Criteria for Significant Improvement**	
	No descriptions of outcomes OR < 20% change in pain rating or functional status	0
	Pain rating with a decrease of 2 or more points or more than 20% reduction OR functional status improvement of more than 20%	1
	Pain rating with decrease of ≥ 2 points AND ≥ 20% change or functional status improvement of ≥ 20%	2
	Pain rating with a decrease of 3 or more points or more than 50% reduction OR functional status improvement with a 50% or 40% reduction in disability score	2
	Significant improvement with pain and function ≥ 50% or 3 points and 40% reduction in disability scores	4
12.	**Description of Drop Out Rate**	
	No description despite reporting of incomplete data or more than 30% withdrawal	0
	Less than 30% withdrawal in one year in any group	1
	Less than 40% withdrawal at 2 years in any group	2
13.	**Similarity of Groups at Baseline for Important Prognostic Indicators**	
	No groups or groups dissimilar with significant influence on outcomes	0
	Groups dissimilar without significant influence on outcomes	1
	Groups similar	2
14.	**Role of Co-Interventions**	
	Dissimilar co-interventions or similar co-interventions in some of the participants	1
	No co-interventions or similar co-interventions in majority of the participants	2
V.	**ASSIGNMENT**	
15.	**Method of Assignment of Participants**	
	Case report/case series or selective assignment based on outcomes or retrospective evaluation based on clinical criteria	1
	Prospective study with inclusion without specific criteria	2
	Retrospective method with inclusion of all participants or random selection of retrospective data	3
	Prospective, well-defined assignment of methodology and inclusion criteria (quasi randomization, matching, stratification, etc.)	4
VI.	**CONFLICTS OF INTEREST**	
16.	**Funding and Sponsorship**	
	Trial included industry employees with or without proper disclosure	-3
	Industry employees involved; high levels of funding with remunerations by industry or an organization funded with conflicts	-3
	Industry or organizational funding with reimbursement of expenses with some involvement or no information available	0
	Industry or organization funding of expenses without involvement	1
	Funding by internal resources only	2
	Governmental funding without conflict such as NIH, NHS, AHRQ	3
TOTAL MAXIMUM		**48**

Source: Manchikanti L, et al. Development of an interventional pain management specific instrument for methodologic quality assessment of nonrandomized studies of interventional techniques. *Pain Physician* 2014; 17:E291-E317 (56).

Appendix 4. *Methodological quality assessment of randomized trials of percutaneous adhesiolysis utilizing Cochrane review criteria.*

	Gerdesmeyer et al, 2013 (17)	Chun-jing et al, 2012 (18)	Manchikanti et al, 2009 (16)	Heavner et al, 1999 (21)	Manchikanti et al, 2009 (15)	Manchikanti et al, 2004 (19)	Veihelmann et al, 2006 (20)
Randomization adequate	Y	Y	Y	U	Y	Y	Y
Concealed treatment allocation	Y	Y	Y	U	Y	Y	Y
Patient blinded	Y	Y	Y	Y	Y	Y	N
Care provider blinded	N	N	N	N	N	N	N
Outcomes assessor blinded	Y	Y	U	Y	Y	Y	N
Drop-out rate described	Y	Y	Y	Y	Y	Y	Y
All randomized participants analyzed in the group	Y	Y	Y	Y	Y	Y	Y
Reports of the study free of suggestion of selective outcome reporting	Y	Y	Y	Y	Y	Y	Y
Groups similar at baseline regarding most important prognostic indicators	Y	Y	Y	Y	Y	Y	Y
Co-intervention avoided or similar in all groups	Y	Y	Y	Y	Y	Y	N
Compliance acceptable in all groups	Y	Y	Y	Y	Y	Y	Y
Time of outcome assessment in all groups similar	Y	Y	Y	Y	Y	Y	Y
Score	11/12	11/12	10/12	9/12	11/12	11/12	10/12

Furlan AD, et al. 2009 updated method guidelines for systematic reviews in the Cochrane Back Review Group. *Spine (Phila Pa 1976)* 2009;34(18):1929-1941 (54)

Appendix 5. *Methodologic quality assessment of randomized trials of percutaneous adhesiolysis interventions utilizing IPM – QRB criteria.*

		Gerdesmeyer et al, 2013 (17)	Chun-jing et al, 2012 (18)	Manchikanti et al, 2009 (16)	Heavner et al, 1999 (21)	Manchikanti et al, 2009 (15)	Manchikanti et al, 2004 (19)	Veihelmann et al, 2006 (20)
I.	Trial design and guidance reporting							
1.	Consort or Spirit	3	0	3	0	3	3	1
II.	Design Factors							
2.	Type and design of trial	3	2	2	2	2	2	3
3.	Setting/physician	1	1	2	2	2	2	1
4.	Imaging	3	3	3	3	3	3	3
5.	Sample size	2	2	3	0	3	2	2
6.	Statistical methodology	1	1	1	1	1	1	1
III.	Patient factors							
7.	Inclusiveness of population	2	2	2	1	2	1	2
8.	Duration of pain	2	1	2	2	2	2	0
9.	Previous treatments	2	2	2	2	2	2	2
10.	Duration of follow-up with appropriate interventions	2	1	2	2	2	2	2
IV.	Outcomes							
11.	Outcomes assessment criteria for significant improvement	4	2	4	0	4	2	0
12.	Analysis of all randomized participants in the groups	2	2	2	1	2	2	1
13.	Description of drop out rate	1	1	1	0	0	1	0
14.	Similarity of groups at baseline for important prognostic indicators	2	2	2	2	2	2	1
15.	Role of co-interventions	0	1	1	1	1	1	1
V.	Randomization							
16.	Method of randomization	2	2	2	0	2	2	1
VI.	Allocation concealment							
17.	Concealed treatment allocation	2	2	2	1	2	1	1
VII.	Blinding							
18.	Patient blinding	1	1	1	1	1	1	1
19.	Care provider blinding	0	0	0	0	0	0	0
20.	Outcome assessor blinding	1	1	0	0	1	1	0
VIII.	Conflicts of interest							
21.	Funding and sponsorship	2	2	2	2	2	2	2
22.	Conflicts of interest	3	3	3	0	3	3	1
Total		41	34	42	23	40	37	25

Source: Manchikanti L, et al. Assessment of methodologic quality of randomized trials of interventional techniques: Development of interventional pain management specific instrument. *Pain Physician* 2014; 17:E263-E290 (55).

Appendix 6. *IPM checklist for assessment of nonrandomized or observational studies of percutaneous and endoscopic adhesiolysis interventions utilizing IPM-QRBNR.*

		Manchikanti et al, 1999 (22)	Gerdesmeyer et al, 2005 (23)	Oh et al, 2014 (24)
I.	Study design and guidance reporting			
1.	Strobe or trend guidance	2	2	0
II.	Design factors			
2.	Study design and type	1	3	1
3.	Setting/physician	2	1	1
4.	Imaging	3	3	3
5.	Sample size	1	0	1
6.	Statistical methodology	2	2	2
III.	Patient factors			
7.	Inclusiveness of population	2	3	4
8.	Duration of pain	2	0	2
9.	Previous treatments	1	2	2
10.	Duration of follow-up with appropriate interventions	3	2	2
IV.	Outcomes			
11.	Outcomes assessment criteria for significant improvement	2	2	2
12.	Description of drop out rate	0	0	0
13.	Similarity of groups at baseline for important prognostic indication	1	0	1
14.	Role of co-interventions	1	1	2
V.	Assignment			
15.	Method of assignment of participants	3	2	3
VI.	Conflicts of interest			
16.	Funding and sponsorship	2	2	2
Total		28/48	25/48	28/48

Manchikanti L, et al. Development of an interventional pain management specific instrument for methodologic quality assessment of nonrandomized studies of interventional techniques. *Pain Physician* 2014; 17:E291-E317 (56).

Appendix 7. *Characteristics of inclusion criteria of randomized trials and observational studies.*

Manuscript Authors	Type of Study	Number of patients	Treatment vs. Comparator	Length of Follow up	Outcome Parameters	Comments
Gerdesmeyer et al, 2013 (17)	RA/PC/P	90	Percutaneous neurolysis 3-day protocol vs subcutaneous injection	12 months	ODI, VAS	High quality, true placebo study showing efficacy in patients with radicular pain with concordant MRI and no absolute stenosis or motor weakness
Chun-jing et al, 2012 (18)	RA/AC/P	76	Ventral epidural vascular catheter with 50-80 ml saline and 10 mg dexamethasone vs 10 mg dexamethasone	6 months	VAS, opioid use	Presence of radiculopathy was an inclusion criterion
Heavner et al, 1999 (21)	RA, AC	59 Epidural fibrosis with radicular pain	3 day adhesiolysis protocol 4 groups: Group I: hypertonic saline plus hyaluronidase Group II: hypertonic saline Group III: isotonic saline (0.9% NaCl) Group IV: isotonic saline plus hyaluronidase	12 months	MPQ VAS for back, right leg, and left leg pain	Purpose of study was to determine if hyaluronidase or hypertonic saline improved the outcome. 29% drop out rate Low back rather than leg pain was the greatest problem. Hyaluronidase was effective with combination with hypertonic saline in reducing frequency of additional procedures. Hyaluronidase did not provide benefit. Hypertonic saline patients required fewer additional treatments than patients treated with normal saline. Maximum VAS scores were improved in between 25% and 60% of patients at 12 months.
Manchikanti et al, 2009 (16)	RA/AC/P	120 Post lumbar surgery syndrome	60 patients receiving 1-day adhesiolysis 60 patients with caudal epidural. Repeat procedures allowed at 3 months based upon initial improvement then deterioration of pain relief to below 50%.	12 months Crossover allowed at 3 months. Of caudal group, 10 were unblinded at 6 months and 33 at 12 months; of the adhesiolysis group, 2 were unblinded prematurely.	NRS, ODI, opioid intake employment/ work status	90% of adhesiolysis group had >50% relief at 3 months and 73% did at 12 months. 35% of caudal group had >50 relief at 3 months and 12% did at 12 months. 77% of adhesiolysis group had >40% improvement in ODI at 12 months compared to 13% of caudal group. Average of 3.5 adhesiolysis procedures/year with an average relief/year of 4½ weeks.
Manchikanti et al, 2009 (15)	RA/AC/P	50 Spinal stenosis	25 patients receiving 1-day adhesiolysis 25 patients with caudal epidural. Repeat procedures allowed at 3 months based upon initial improvement then deterioration of pain relief to below 50%.	12 months Of caudal group, 18 were unblinded prematurely. None of the adhesiolysis group were unblinded prematurely.	NRS, ODI, opioid intake, employment/ work status	76% of adhesiolysis patients had significant relief at one year.

Manchikanti et al, 2004 (19)	RA/AC	75 Low back pain without response to epidural injection and no facet disease Between 64% and 72% patients had prior lumbar surgery; between 4% and 20% had spinal stenosis	25 caudal epidural steroid injection 25 1-day adhesiolysis with normal saline 25 1-day adhesiolysis with hypertonic saline Patients averaged 2.1 to 2.7 procedures	12 months Unblinding at 3 or 6 months	VAS, ODI, work status, opioid intake, range of motion, psychological evaluation by P3	72% of hypertonic saline and 60% of normal saline patients had >50% relief at 12 months, versus 0% of caudal injections. 18 of the caudal group were unblinded by 6 months.
Veihelmann et al, 2006 (20)	RA/AC	99	One day adhesiolysis with 10% saline versus physical therapy	3, 6, 12 months	VAS, ODI, GHS	Mean improvement of the treated group was >50% in VAS and >40% in ODI at 3, 6 and 12 months. Treatment group had ~10% relief.
Manchikanti et al, 1999 (22)	RE	150	One-day vs Two-day adhesiolysis 3-day adhesiolysis results were obtained from a different study (19) Up to four procedures in one year	3, 6, 12 months	Pain relief	Study showing equivalency of one-day and three-day adhesiolysis. Does not have functional evaluation
Gerdesmeyer et al, 2005 (23)	Prospective	61	Three day adhesiolysis	6 months	ODI, McNabb Score	Preliminary study for subsequent randomized study
Oh et al, 2014 (24)	Retrospective review	303 Single level disc disease with or without radicular pain without prior surgery or spinal stenosis	One day adhesiolysis with the catheter positioned either ventrally or dorsally	6 months	VAS back and leg; Odom's criteria	Study evaluated whether catheter positioning either ventrally or dorsally influenced outcomes

RA = randomized; PC = Placebo control; AC = Active-control; P = Prospective; RE =Retrospective; VAS = Visual analog scale; ODI = Oswestry Disability Index; NRS = Numeric Rating Scale; P-3 = Pain Patient Profile; MPQ = McGill Pain Questionnaire; GHS=Gerbershagen score

Appendix 8. *Study characteristics of randomized controlled trials and observational studies assessing percutaneous adhesiolysis.*

Study Study Characteristic Methodological Quality Scoring	Number of Patients & Selection Criteria	Interventions/Control	Outcome measures	Time of Measurement	Results	Weaknesses	Strengths	Conclusions
Gerdesmeyer et al, 2013 (17) Randomized, Placebo controlled. Cochrane 11/12 IPM-QRB 41/48	90 Patients with radicular pain, positive Leségue sign, concordant MRI and no muscle weakness	Adhesiolysis 3-day protocol with ventrolateral catheter placement	ODI/VAS	3, 6 and 12 months	28/31 of the treated group had a >50% improvement in ODI at 12 months, vs 9/26 of the control 29/31 of the treated group have >50% improvement in VAS at 12 months, vs 18/26 of control	Did not measure change in opioid consumption	Well designed prospective, high quality placebo controlled study with functionality as primary outcome	Documents efficacy versus placebo, with improvements in function and pain.
Chun-jing et al, 2012 (18) Randomized, active control Cochrane 11/12 IPM-QRB 34/48	92 patients with pain and radiculopathy 6 months after surgery for disc herniation 76 patients were evaluated	Vascular catheter with guidewire was passed to ventral epidural space. 50-90 ml and 10 mg of dexamethasone were injected. Control got 10 mg dexamethasone only	VAS, Opioid use; MacNab criteria	1 week, one month,6 months	3.24 mean decrease in VAS at 6 months in treated group, vs .82 in control. Patients in treated group who did not have improvement in dye flow did not have clinical improvement	Lateral views only were obtained; no measure of function	High quality study	Adhesiolysis is effective using the vascular catheter with ventral placement of the catheter. Improvements in dye flow are necessary for good clinical outcomes.
Manchikanti et al, 2009 (16) Randomized Active Control Cochrane 11/12 IPM-QRB 34/48	120 Post lumbar surgery syndrome	Adhesiolysis with 10% saline vs S3 caudal injection with 0.9% saline	NRS, ODI, Employment status, opioid use	3, 6 and 12 months	~70% of adhesiolysis procedures had >50% relief and also >40% ODI improvement at 12 months, vs ~12% of caudal. Average of 3.5 adhesiolysis procedures vs. 2.2 caudal injection	33 of 60 in the control group were lost to follow up at 12 months, vs 2 in the adhesiolysis. 43 control were unblinded prematurely vs 2 in the control	High quality equivalency study.	Adhesiolysis, one day, repeated up to 4 times a year, is effective in providing decreased pain and increased function in the post lumbar surgery population.
Heavner et al, 1999 (21) Randomized active control IPM-QRB 23/48	83	Adhesiolysis in four groups, 0.9% saline, 10% saline, with or without hyaluronidase	VAS, McGill Pain Questionnaire	3, 6, 12 months	No significant difference between the four groups. Adhesiolysis did provide pain relief in about 50% of subjects. Most subjects require more than one procedure	Successful pain relief was 10/100 reduction in VAS. 24 of 83 patients were removed from study Ventral	Showed only moderate additional benefit from either 10% saline or hyaluronidase	Moderate quality study comparing 4 treatment options Reduced additional procedures.
Manchikanti et al, 2009 (15) Randomized Active Control IPM-QRB 40/48	50 spinal stenosis with low back and leg pain	Adhesiolysis with 10% saline vs S3 caudal injection with 0.9% saline	NRS, ODI, Employment status, opioid use	3, 6, 12 months	76% of adhesiolysis procedures had >50% relief and also >40% ODI improvement at 12 months, vs 4% of caudal.	8 of 25 in the control group were lost to follow up at 12 months, vs 0 in the adhesiolysis. 18 control were unblinded prematurely vs 0 in the control	High quality equivalency study.	Adhesiolysis, one day, repeated up to 3.5 times a year, is effective in providing decreased pain and increased function in the spinal stenosis population.
Manchikanti et al, 2004 (19) Randomized Active	75	One-day adhesiolysis with either 0.9% or 10% saline vs	VAS, ODI, work status, opioid intake, range of motion	3.6.12 months	72% of adhesiolysis and hypertonic neurolysis and 60% of 0.9% saline	18 of the ESI group were unblinded by 6 months.	Comparison of hypertonic and normal saline vs epidural	High quality RCT showing that adhesiolysis provides

Study Study Characteristic Methodological Quality Scoring	Number of Patients & Selection Criteria	Interventions/Control	Outcome measures	Time of Measurement	Results	Weaknesses	Strengths	Conclusions
Control Cochrane 11/12 IPM-QRB 37/48		epidural steroid injection.	measurement, and P-3		adhesiolysis compared to 0% of epidural group had >50% relief at12-months	Repeat procedures allowed based upon response to previous procedures, rather than examining one injection only.	steroid injection	significant relief regardless of whether normal saline or hypertonic saline is used.
Veihelmann et al, 2004 (20) Randomized prospective study with active control Cochrane 11/12 IPM-QRB 25/48	99 patients with radicular pain with concordant imaging findings	One day adhesiolysis with 10% saline vs. physical therapy	VAS, ODI, GHS	3, 6, 12 months	Mean improvement of the treated group was >50% in VAS and >40% in ODI at 3, 6 and 12 months. Treatment group had ~10% relief.	25 patients in the control group were lost t follow up, 10 lost to follow up, 12 crossing over to adhesiolysis, 10 having surgery	Moderate quality comparative effectiveness study	Adhesiolysis is superior to physical therapy in treating persistent back and leg pain with concordant imaging findings.
Manchikanti et al, 1999 (22) Retrospective observational IPM-QRBNR 28/48	150 patients selected from 532 who had had adhesiolysis.	One day vs 2 day adhesiolysis with hypertonic saline Compared to Racz 1999 for 3 day adhesiolysis	Pain relief of > 50%	3, 6, 12 months	No difference between one-day, two-day or three-day adhesiolysis Repeat procedures showed significant (>50%) relief with longer duration.	Retrospective study, with one leg obtained from an unrelated study.	Large scale study with	Moderate quality study showing that one day adhesiolysis is as effective as three-day.
Gerdesmeyer et al, 2006 (23) Prospective observational IPM-QRBNR 25/48	61 patients with radiculopathy	Ventral three-day adhesiolysis	ODI, MacNab scale	3, 6 months	ODI was reduced >50% at 3 and 6 months	Mixture of multiple etiologies No control group	Prospective, tightly controlled study	Moderate quality study documenting that replication of pain leads to functional improvement in adhesiolysis.
Oh et al, 2014 (24) Retrospective IPM-QRBNR 28/48	303 patients with one-level disc disease with low back and or leg pain. No previous surgery or stenosis	One-day adhesiolysis with either ventral or dorsal catheter placement	VAS, Odom's criteria	1, 3, 6 months	VAS for leg pain was significantly greater at 3 and 6 months for ventral group. Both ventral and dorsal placement showed >50% reduction of back and leg pain	Retrospective study with no control.	Large scale, single center study	Ventral placement appears to provide better outcomes.

VAS= Visual Analog Scale; NRS=Numeric Rating Scale; ODI=Oswestry Disability Index; RCT = randomized controlled trial; GHS=Gerbershagen score